nanoker

CERAMIC TECHNICAL DATA SHEET

NC3

Ce-stabilized ATZ Nanocomposite

KEY PROPERTIES

- Ultra-high toughness
- High strength
- Elastic-plastic behavior
- No ageing (no low temperature degradation)
- High wear resistance under sliding conditions
- Low friction coefficient against metals

APPLICATIONS

- Structural parts for machinery
- Welding pins
- Tooling components
- Cutting blades
- Shock-resistant parts

MAIN PROPERTIES

Property	Value	Unit
Density	5.61	g/cm^3
Flexural strength	900	MPa
Flexural strength (HIP)	996	MPa
Fracture Toughness	13.8	MPa·√m
Hardness	13.1	GPa
Young Modulus	254	GPa
CTE (20–500ºC)	10.8	$x10^{-6}$ K^{-1}
Thermal Conductivity	8.36	W/m·K
Electrical Resistivity	$>10^{11}$	Ω·cm

** All properties measured at 20ºC unless otherwise stated.*
*** Measured in a wall thickness of 2 mm*

OPERATING CONDITIONS

Max working temp (air)	1200	ºC
Recomended temp	<1100	ºC
Continuous temp	1000	ºC

ELECTRICAL BEHAVIOR

Insulator

MATERIAL INFORMATION

Material type	Ce-stabilized ATZ nanocomposite
Structure	Advanced Nanocomposite ceramic
Key feature	Extreme toughness
Machinability	Medium
Color	White

Local overheating may occur at electrical contact points due to current concentration and geometry effects.

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